

CHAPTER FOUR

PROBABILITY AND PROBABILITY DISTRIBUTIONS

4.1. PROBABILITY THEORY

❖ Probability, Experiment, Outcomes and Event

Probability is a numerical measure of the likelihood that an event will occur. Thus, probabilities can be *used as measures of the degree of uncertainty* associated with the occurrence of different events. If probabilities are available, we can determine the likelihood of each event occurring. Probability values are always assigned on a scale from 0 to 1. A probability near zero indicates an event is unlikely to occur; a probability near 1 indicates an event is almost certain to occur. Other probabilities between 0 and 1 represent degrees of likelihood that an event will occur.

Experiment: In discussing probability, we define an experiment as a *process that generates* well defined outcomes. On any single repetition of an experiment, one and only one of the possible experimental outcomes will occur.

Outcome is the result of an experiment. An experimental outcome is also called a **sample point** to identify it as an element of the sample space. The sample space for an experiment is the set of all experimental outcomes.

Event: an event is a collection of sample points.

Probability of An Event: The probability of any event is equal to the sum of the probabilities of the sample points in the event. Using this definition, we calculate the probability of a particular event by adding the probabilities of the sample points (experimental outcomes) that make up the event.

Example:

Experiment

Toss a coin
Select a part for inspection
Conduct a sales call
Roll a die
Play a football game

Experimental Outcomes

Head, tail
Defective, non-defective
Purchase, no purchase
1, 2, 3, 4, 5, 6
Win, lose, tie

4.3 APPROACHES IN PROBABILITY

4.3.1 Objective Probability

4.3.1.1 Classic Probability: It is probability based on the symmetry of games of chance or similar situations. This probability is based on the idea that certain occurrences are equally likely. E.g. The numbers 1, 2, 3, 4, 5, and 6 on fair die are equally likely to occur i.e they do have equal chance of occurrence.

4.3.1.2 Long-term Relative Frequency Probability

The probability of an event happening in the long-term is determined by observing what fraction of the time similar events happened in the past. We often think of a probability in terms of the percentage of the time the event would occur in many repetition of the experiment. Suppose that A is an event that might occur when a particular experiment is performed then the probability that the event A will occur, $P(A)$, can be interpreted to be the number that would be approached by the relative frequency of the event A, if we perform the experiment an indefinitely large number of times.

e.g. when we say that the probability of obtaining a head when we toss a coin is 0.5 we are saying that, when we repeatedly toss the coin an indefinitely large number of times, we will obtain a head 50% of the repetition.

In terms of formula

Probability of an event happening = $\frac{\text{Number of times occurred in past}}{\text{Total number of observation}}$

If a truck operator experienced 5 accidents out of 50 truck last year, then the probability that a truck will have an accident next year can be $5/50 = 0.10$

4.3.2 Subjective Probability

When there is no past experience or little on which to base a probability, personal judgment, experience, intuition or expertise or any other subjective evaluation criteria will be applied to estimating or assigning probability. This probability is subjective probability.

It is also called personal probability. Unlike objective probability one person's subjective probability may very well different from another person's subjective probability of the same event. eg. A physician assessing the probability of a patient's recovery and an expert in the national bank assessing probability of currency devaluation are both making a personal judgment based on what they know and feel about the situation and other group of physicians or experts will arrive with different probability, though both can employ identical techniques or approaches and information.

Both classic and long-term relative frequency probabilities are objective in the sense that no personal judgment is involved. Whatever the kind of probability involved (i.e. subjective or objective), the same set of mathematical rules holds for manipulating and analyzing probability.

4.4 SAMPLE SPACE AND SAMPLE SPACE OUTCOME

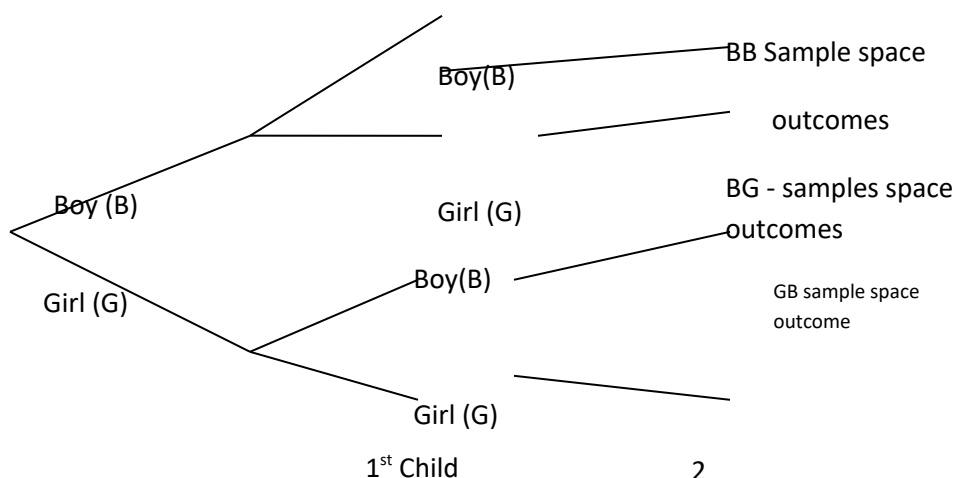
In order to calculate and interpreter probabilities it is important to understand and use the idea of sample space. *The sample space of an experiment* is the set of all of the distinct possible

outcomes of the experiment. Each distinct outcome is called sample space outcome or sample point or elementary event.

Example 1: A newly married couple plans to have two children. Naturally, they are curious about whether their children will be boys or girls. Therefore, we consider the experiment of having two children. In order to find the sample spaces of this experiment, of having two children, we let 'B' denote that child is a boy and 'G' denotes that child is a girl.

This experiment is a two-step process i.e having the first child, which could be a boy or a girl and having the second child, which could also be either a boy or a girl.

This can be constructed by a tree diagram. Each branch of the tree leads us to a distinct sample space outcome.



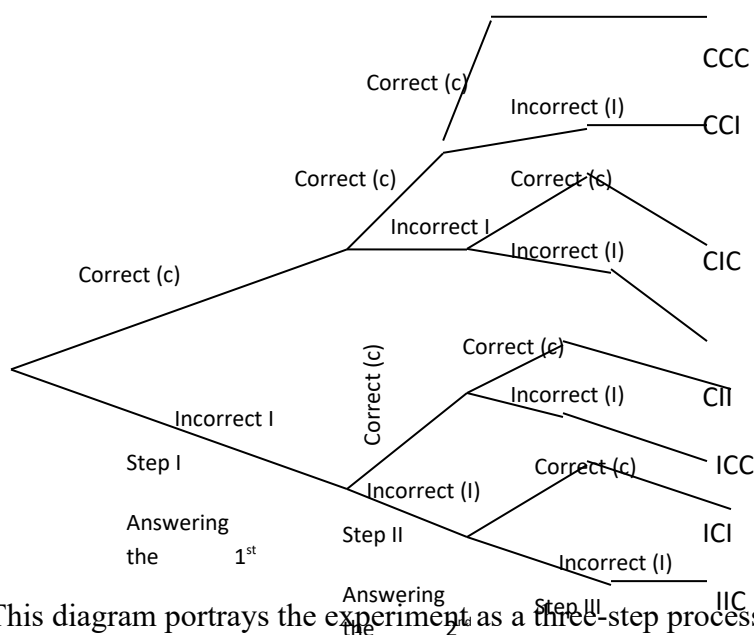
We see that there are four sample space outcomes. Therefore the sample space of the distinct samples space outcomes is BB BG GB GG.

In order to consider the probabilities of these outcomes, suppose that boys and girls are equally likely each time a child is born. This says that each of the sample space outcomes is equally likely. i.e. $P(BB) = P(BG) = P(GB) = P(GG) = \frac{1}{4}$. This says that there is a 25% chance that each of these outcomes will occur. Since we are certain that there is no other option or combination remaining, the probability that the couple will have any one of the sample space outcomes is one. i.e. $P(BB) + P(BG) + P(GB) + P(GG) = 1$

Notice that these probabilities sum one i.e the sum of the probabilities of all sample space outcomes is one. Therefore the sample space (that is, the set of all of the distinct sample space outcomes) is BB, BG, GB, GG

Example 2: A student takes a quiz that consists of three true or false questions. If we consider our experiment to be answering the three questions, each question can be answered correctly or incorrectly.

Let C denote answering a question correctly and I denote answering a question incorrectly. Then we can depict a tree diagram of the sample space outcome for the experiment.



This diagram portrays the experiment as a three-step process

Step I – answering the 1st question (Correctly or Incorrectly) (C or I)

Step II – answering the 2nd question (Correctly or incorrectly).

Step III – answering the 3rd question (Correctly or incorrectly).

The tree diagram has eight different branches and the eight distinct sample space outcomes are listed at the end of the branches. We see the sample space is

CCC CCI CIC CII

ICC ICI IIC III

Now suppose that the student was totally unprepared for the test, and has to blindly guess the answer to each question that is the student has a 50-50 chance or 0.5 probability of correctly answering each question. This means that each of the eight sample space outcomes is equally likely to occur.

i.e $P(ccc) = P(ccI) = \dots = P(III) = 1/8$

Here also the sum of the probabilities of the sample space out comes is one.

In General the sum of the probabilities of all the sample space is equal to 1.

Finding Probabilities by using Sample Space

If all of the sample space outcomes are equally likely, then the probability that an event will occur is equal to the ratio: **The number of sample space outcomes that correspond to the event to the total number of sample space outcomes.**

Consider the couple planning to have two children to find the probability of two boys. First we have to find the sample space outcome corresponding to the event of having the first child a boy and the second child also a boy.

There is only one sample space outcome corresponding to this event i.e. BB so the probability

will be: $\frac{1}{4} = 0.25$ the probability that the couple will have a boy and a girl is similarly calculated by first identifying the sample space outcomes corresponding to the event of having a boy and a

girl. The sample space outcomes are BG and GB. So the probability will be $\frac{2}{4} = 0.5$

Check Your Progress – 1

1. Suppose that a couple will have three children. Letting B denote a boy and G denote a girl.
 - a) Draw a tree diagram depicting the sample space outcome for this experiment.
 - b) list the sample space outcomes that correspond to each of the following events.
 - 1) All three children will have the same gender
 - 2) Exactly two of the three children will be girls.
 - 3) Exactly one of the three children will be a girl.
 - 4) None of the tree children will be a girl.
2. Four people will enter an automobile show Room and each will either purchase a car (P) or will not purchase a car (N)
 - a) Draw a tree diagram depicting the sample space of all possible purchase decision that could potentially be made by the four people.
 - b) List the sample space out comes that correspond to each of the following events.
 - 1) Exactly three people will purchase a car
 - 2) Two or fewer will purchase a car
 - 3) One or more people will purchase a car
 - 4) All four people will make the same purchase decision

Often time it may be practically impossible to list all possible sample space outcomes of an experiment. Under such circumstances we can find the probability of an event by identifying the number of sample space outcomes /without listing/ corresponding to the event.

Example - Suppose that 650,000 of 1,000,000 households in Addis subscribe to a newspaper called Addis Zemen, and consider randomly selecting one of the Households in this city. That is consider selecting one household & giving each and every household in the city the same chance of being selected. Let A be the event that the randomly selected household subscribes to the Addis Zemen. Then since the sample space of this experiment consists of 1,000,000 equally likely sample space outcomes (households). It follows that

$$P(A) = \frac{\text{the number of Households that subscribe to the Addis Zemen}}{\text{The total number of households in the city}} \\ = \frac{650,000}{1,000,000} = 0.65$$

Now also suppose that 500,000 households in the city subscribe to the Ethiopian Herald (H) and further suppose that 250,000 households subscribe to both the newspapers.

We consider randomly selecting one household in the city, and we define the following events

A = The randomly selected household subscribes to the Addis Zemen.

$\bar{A}$ = The randomly selected, household does not subscribe to the Addis Zemen.

H = The randomly selected household subscribes to the Ethiopian Herald.

$\bar{H}$ = The randomly selected household does not subscribe the Herald.

Using the notation $A \cap H$ to denote both A & H we also define.

$A \cap H$ = The randomly selected household subscribes both to Addis Zemen & Herald.

Since 650,000 of the 1,000,000 households subscribe to the Addis Zemen (that is correspond to the event Occurring). Then 350,000 households do not subscribe to Zemen ($\bar{A}$) i.e. $1,000,000 - 650,000$. Similarly since 500,000 households subscribe to Herald (H) 500,000 households do not subscribe to herald ($\bar{H}$)

Next consider the events $A \cap \bar{H}$ = the randomly selected household subscribes to Zemen and does not subscribe to Herald; $\bar{A} \cap H$ = the randomly selected household does not subscribe to Zemen and does subscribe to Herald.

A summary of the number of households corresponding to the events A, $\bar{A}$, H, $\bar{H}$ and $A \cap H$

Events	Subscribe to Herald	Does not subscribe to Herald	Total
Subscribe to Addis Zemen	250,000		650,000
Does not subscribe to Addis			350,000

Zemen			
Total	500,000	500,000	1,000,000

Define the event $\bar{A} \cap \bar{H}$,

$\bar{A} \cap \bar{H}$ = the randomly selected household does not subscribe to both newspaper.

Since 650,000 households subscribe to the Addis Zemen (A) and 250,000 households subscribe to both Zemen and Herald ($A \cap H$) it follows that $650,000 - 250,000 = 400,000$ households subscribe to Addis Zemen but do not subscribe to Herald, ($A \cap \bar{H}$). This subtraction is illustrated in the table below.

By similar logic

- $500,000 - 250,000 = 250,000$ households do not subscribe to Addis Zemen but do subscribe to Herald ($\bar{A} \cap H$)
- $350,000 - 250,000 = 100,000$ households do not subscribe the Addis Zemen and also do not subscribe the Herald ($\bar{A} \cap \bar{H}$)
- Subtracting to find the number of households corresponding to the events.
- $A \cap H, A \cap \bar{H},$

Event	H	$\bar{H}$	
A	250,000	650,000-250,000	650,000
$\bar{A}$			350,000
Total	500,000	500,000	1,000,000

e. $(\bar{A} \cap H) = 500,000 - 250,000$

$= 250,000$

f. $(\bar{A} \cap \bar{H}) = 350,000 - 250,000$

$= 100,000$

A contingency table summarizing subscription data for Addis Zemen and Herald

Event	Subscribe to Herald (H)	Does not Subscribe to Herald ($\bar{H}$)	Total
Subscribe to Addis Zemen (A)	250,000	400,000	650,000
Does not subscribe to Addis	250,000	100,000	350,000

Zemen ($\bar{A}$)			
Total	500,000	500,000	1,000,000

Now since we will randomly select one household (making all the households equally likely to be chosen), the probability of any of the previously defined events is the ratio of the number of households corresponding to the event's occurrence to the total number of households in the city.

Therefore

$$P(A) = \frac{650,000}{1,000,000} = 0.65$$

$$P(H) = \frac{500,000}{1,000,000} = 0.5$$

$$P(A \cap H) = \frac{250,000}{1,000,000} = 0.25$$

Next letting AUH denote either A or H, we consider finding the probability of the event

AUH = the randomly selected household subscribes to either the Addis Zemen or Herald. (i.e. subscribe to at least one of the two newspapers).

We see that the households subscribing to either Addis Zemen or Herald:

- a) The 400,000 households that subscribe to only Addis Zemen, $A \cap \bar{H}$
- b) The 250,000 households that subscribe to only the Herald, $\bar{A} \cap H$ and
- c) The 250,000 households that subscribes to both Addis Zemen and Herald, $A \cap H$. Therefore since a total of 900,000 households subscribe to either the Addis Zemen or Herald it follows:
 - $P(AUH) = \frac{900,000}{1,000,000} = 0.9$

i.e 90% of the households in the city subscribe to either Addis Zemen or Herald.

Notice that $P(AUH) = 0.9$ does not equal $P(A) + P(H) = 0.65 + 0.5 = 1.15$

Logically the reason for this is that both $P(A) = 0.65$ and $P(H) = 0.5$ count the 25% of the households that subscribe to both newspapers. Therefore; the sum of $P(A)$ and $P(H)$ counts this 25% of the households once too often.

It follows that if we subtract $P(A \cap H) = 0.25$ from the sum of $P(A)$ and $P(H)$ then we will obtain

$$P(A \cup H) \text{ i.e } P(A \cup H) = P(A) + P(H) - P(A \cap H) \\ = 0.65 + 0.5 - 0.25 = 0.90$$

The intersection and union of Two events.

Given two events A&B

- 1) The Intersection of A&B is the event consisting of the sample space outcomes belonging to both A&B, denoted by $A \cap B$. Furthermore; $P(A \cap B)$ denotes the probability that Both A&B will simultaneously occur.
- 2) The union of A&B is the event consisting of sample space outcomes belonging to either A or B. The union is denoted $A \cup B$ Furthermore; $P(A \cup B)$ denotes the probability that either A or B will occur.

2.5 PROBABILITY RULES

2.5.1 The Addition Rule

2.5.1.1 Addition Rule for two Dependent Events

Let A and B be events then the probability that either A or B will occur is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

2.5.1.2 Addition Rule for Two Mutually Exclusive Events

Two events are said mutually exclusive if they have no sample space outcomes in common. In this case, the event A&B cannot occur simultaneously and thus. $P(A \cap B) = 0$.

Let A&B are mutually exclusive events then, the probability that either A or B will occur is

$$P(A \cup B) = P(A) + P(B)$$

Example - consider randomly selecting a card from a standard deck of 52 playing cards and define the events. J, a randomly drawn card is Jack; Q, a randomly drawn card is Queen; and K, a randomly drawn card is a king. Since there are 4 Jacks, 4 Queens and 4 Kings in the deck, $P(Q) = \frac{4}{52}$, $P(K) = \frac{4}{52}$, $P(J) = \frac{4}{52}$.

Since there is no card that is both a J & Q the event J and Q are mutually exclusive and thus

$P(J \cap Q) = 0$ it follows that the probability that the randomly selected card is either J or Q is

$$P(J \cup Q) = P(J) + P(Q) = \frac{4}{52} + \frac{4}{52} = \frac{2}{13}$$

2.5.1.3 The Addition Rule for N mutually exclusive events.

The event $A_1, A_2, \dots, A_n$ are mutually exclusive if no two of the events have any sample space out come in common. In this case no two of the events can occur simultaneously and

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$$

Example $P(J \cup Q \cup K \cup \text{nine}) = P(J) + P(Q) + P(K) + P(\text{nine})$

$$= \frac{4}{52} + \frac{4}{52} + \frac{4}{52} + \frac{4}{52} = \frac{16}{52}$$

2.6 THE COMPLEMENT OF AN EVENT

Given an event A, the complement of A is the event consisting of all sample space outcomes that do not correspond to the occurrence of A.

The complement of A is denoted by $\bar{A}$

Furthermore $P(\bar{A})$ denotes the probability that A will not occur.

In any probability situation, either an event A or its complement A must occur.

Therefore we have

$$P(A) + P(\bar{A}) = 1 \text{ This implies } P(\bar{A}) = 1 - P(A)$$

Example – If team A and B are playing for a final cup we can say that the events that team A will win is complement to the event that B will win. i.e., if A wins B will lose. Under no circumstance that A will win and loses at the same time winning and losing are mutually exclusive.

4.7 CONDITIONAL PROBABILITY AND INDEPENDENCE

4.7.1 Conditional Probability

Probability is conditional upon information. We may define the probability of event A conditional upon the occurrence of event B.

If we think about two adjacent rooms, R_1 and R_2 , the probability that R_1 will be caught by fire is highly conditional on the probability of the other room.

Example 1. Suppose that we randomly select a household, and that the chosen household reports it subscribes to Herald. Given this new information we wish to find the probability that this household subscribes to Addis Zemen. The new probability is called a conditional probability.

The probability of the event A, given the condition that the event H has occurred, is written

$P(A/H)$ = the probability of A given H. We often refer to such a probability as the conditional probability of A given H .

In order to find the conditional probability that a household subscribes to Addis Zemen given that it subscribes to Herald we know that we are considering one of 500,000 households. Since 250,000 of these 500,000 Herald subscribers also subscribe to Addis Zemen we have

$$P(A/H) = \frac{250,000}{500,000} = 0.5$$

i.e 50% of the Herald subscribers also subscribe to Addis Zemen:

Example 2. Next suppose that we randomly select another household from the 1,000,000 households and suppose that this newly chosen household reports that it subscribes to Addis Zemen

Now find the probability that this household subscribes to Herald

$$P(H/A)$$

$$= \frac{250,000}{650,000} = 0.3846$$

This says that the probability that the randomly selected household subscribes to herald given that the household subscribes to Addis Zemen is 0.3846. ie., 38.46% of Addis Zemen subscribers also subscribe to Herald.

We have

$$P(A) = \frac{650,000}{1,000,000} = 0.65$$

$$P(A \cap H) = \frac{250,000}{1,000,000} = 0.25$$

$$P(H/A) = \frac{250,000}{650,000} = 0.3846$$

$$P(H) = \frac{500,000}{1,000,000} = 0.5$$

$$P(A/H) = \frac{250,000}{500,000} = 0.5$$

If we divide both the numerator and denominator of each conditional probability by 1,000,000

$$P(A/H) = \frac{250,000}{500,000} = \frac{250,000/1,000,000}{500,000/1,000,000} = \frac{P(A \cap H)}{P(H)}$$

$$P(H/A) = \frac{250,000}{650,000} = \frac{250,000/1,000,000}{650,000/1,000,000} = \frac{P(A \cap H)}{P(A)}$$

We express these conditional probabilities in terms of $P(A)$, $P(H)$ and $P(A \cap H)$

Given the sample space outcomes are equally likely.

$$P(A/H) = \frac{P(A \cap H)}{P(H)}, \text{ then } P(A \cap H) = P(H) P(A/H), \text{ by simple cross multiplication}$$

$$P(H/A) = \frac{P(A \cap H)}{P(A)} = \text{then } P(A \cap H) = P(A) P(H/A)$$

The General Multiplication Rule

(two ways to calculate $P(A \cap H)$)

$$\boxed{P(A \cap H) = P(A) P(H/A) = P(H) P(A/H)}$$

Example 1. In a firm 20% of the employees have an accounting background, while 5% of the employees are executives and have an accounting background. If an employee has accounting background, what is the probability that the employee is an executive?

Let us define the events

E, an employee is an executive and

A, an employee has an accounting background

$$P(A) = 0.2$$

$$P(A \cap E) = 0.05, \text{ then}$$

$$P(E/A) = \frac{P(A \cap E)}{P(A)} = \frac{0.05}{0.2} = \underline{0.25}$$

Example 2. A contractor is bidding for two projects with Co. A and Co. B. The contractor estimates that the probability of obtaining the project with Co. A is 0.45. He also feels that if he should get the project with Co. A then there is a 0.90 probability that Co. B will also give him the project. What are the contractor's chances of getting both projects?

Solution: We are given

$$P(A) = 0.45$$

$P(B/A) = 0.90$ and we are looking for $P(A \cap B)$, which is the probability that both A and B will occur. From the equation we have

$$P(A \cap B) = P(B/A) P(A) = 0.9 \times 0.45 = 0.405$$

Check Your Progress –2

21% of the executive in a large firm are at the top salary level. It is further known that 40% of all the executives at the firm are women. Also 6.4% of all executives are women and are at the top salary level. Recently among executives at the firm arose a question as to whether there is any evidence of salary inequality. Check.

Clue. To solve this problem, pose a question in terms of probabilities. I.e., ask whether the probability that the executive will be at the top salary level given the executive is a woman. If this probability is less than 16% (the average) you can conclude that salary inequity does exist because of gender.

2.7.2 Statistical Independence

If the occurrence of events A and B have nothing to do with each other, then we know that A and B are independent events.

i.e the probability of occurrence of A will not influence the probability of occurrence of B.

This implies that

$$P(A/B) = P(A) \text{ and that}$$

$$\underline{P(B/A) = P(B)}$$

Furthermore; the general multiplication rule tells us that, for any two events A and B we can say that

$$P(A \cap B) = P(A) P(B/A) \text{ therefore; if } P(B/A) = P(B) \text{ it follows that}$$

$$P(A \cap B) = P(A) P(B)$$

This is called the multiplication rule for two independent events.

However, if the probability of an event is influenced by whether or not another event occurs, we say the two events are dependent.

eg. Define the events C and P as follows

C= your favorite college football team will win its first match next season.

P= Your favorite professional football team will win its first match next season.

Suppose that you believe that for next season $p(c) = 0.6$ and $p(p) = 0.6$ then since the outcomes of a college football games and a professional football game would probably have nothing to do with each other, it is reasonable to assume that C and P are independent events.

It follows that : Both your favorite teams will win their first match next season,

$$P(C \cap P) = p(c) \cdot p(p) = 0.6(0.6) = 0.36$$

When two events are independent, neither are their complements.

2.7.3 Independent and Mutually Exclusive Events

When two events are mutually exclusive they are not independent. In fact they are very dependent events in the sense that if one happens the other cannot happen. The intersection of two mutually exclusive events is zero but the probability of the intersection of two independent events is not zero. It is equal to the product of the probabilities of the separate events.

2.7.3.1 The multiplication rule for N independent events

The events $A_1, A_2, \dots, A_n$ are independent events if the occurrence of these events have nothing to do with each other. if events $A_1, A_2, \dots, A_n$ are independent events, then

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) P(A_2) \dots P(A_n)$$

Example 1. An electronic device has four independent components C_1, C_2, C_3, C_4 , with a reliability of 0.85 each. The device works only if all four components are functional.

What is the probability that the device will work when needed?

$$\begin{aligned} P(\text{the device will work}) &= P(\text{all components will work}) = P(c_1, nc_2, nc_3, nc_4) \\ &= p(C_1) p(C_2) p(C_3) p(C_4) \\ &= 0.85 \times 0.85 \times 0.85 \times 0.85 \\ &= 0.85 \times 0.85 \times 0.85 \times 0.85 = \underline{0.522} \end{aligned}$$

Example 2. The rate of defects in corks of wine is 0.75. Assuming independence, if four bottles are opened (B_1, B_2, B_3, B_4), what is probability that four corks are defective.

$$P(\text{all 4 are defective}) = P(B_1 \cap B_2 \cap B_3 \cap B_4) = P(B_1) P(B_2) P(B_3) P(B_4) \\ = 0.75 \times 0.75 \times 0.75 \times 0.75 = 0.316$$

2.7.3.2 Union rule

The union of several independent events is the event that at least one of the events happens.

The probability of the union of several independent events $A_1, A_2, \dots, A_n$ is

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = 1 - P(\bar{A}_1) P(\bar{A}_2) \dots P(\bar{A}_n)$$

Example 1: A device similar to the above one has three components, but the device works as long as at least one of the components is functional. The reliability of the components are 0.96, 0.91 and 0.80 what is the probability that the device will work when needed?

$$P(\text{The device will work}) = p(\text{at least one will work}) = 1 - p(\text{all will fail}) = 1 - p(\bar{A}_1)P(\bar{A}_2)P(\bar{A}_3) = 1 - (0.04)(0.09)(0.02)$$

Example 3: In the developing world a woman's odds of dying from problems related to pregnancy is 1 in 51. If three women are pregnant what is the probability that at least one will die

$$p(\text{at least one will die}) = 1 - p(\text{all will survive}) \\ 1 - (50/51)^3 = 0.0577$$

2.2. Probability Distribution

Basic definitions and terms

1. **Random variable:** is a variable whose results are determined by chance.

For example, suppose we toss a coin twice (or toss two coins) and call heads on the toss as success. The number of successes then could be 0, 1 or 2. Which particular value occurs is a matter of chance. So the number of heads is a random variable. Let's give this random variable the general name X . Now note that there is a distinction between the name of the random variable, X , and the values 0, 1, or 2 which the random variable could have. If we are interested in the probability that X takes on the value, we could express this as

$$P(\text{No. of heads}=1) \text{ or}$$

$$P(X=1)$$

✓ Random variable is the one that can take a number of values, but it is difficult to know which value does take until some experiment is performed.

Example: -

➤ No of absentees

➤ No of heads

✓ There are two types of random variable. These are

i. **Discrete Random variable:** is a random variable that is allowed to take only a limited number of values, which can be listed. It can assume only certain clearly separated values.

E.g. When two coins are tossed, the random variable "number of heads" can take on only the value 0, 1, or 2. These values are separated or discrete and the variable "number of heads" is

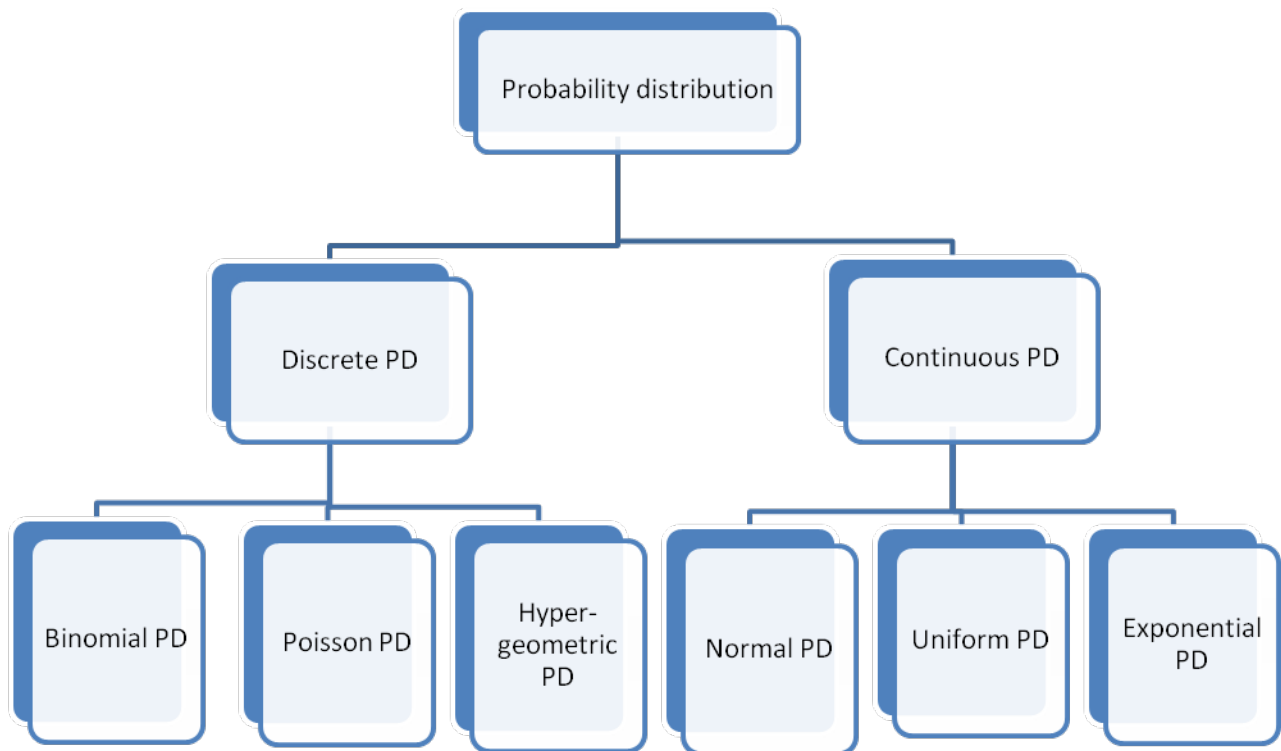
called a discrete random variable.

- ii. **Continuous Random variable:** If a random variable could have any value in some interval of value, it is called continuous random variable. It is allowed to assume any value within a given range.

E.g. A continuous random variable could have any value from, say, 0 through 5, would graph as a continuous straight line.

2. **Probability distribution:** is a complete list of all possible outcomes that may result from an experiment and the probability associated with each outcome. There are two types of probability distribution. They are;

- Discrete probability distribution: is a set of possible values of a discrete random variable along with the probability of the occurrence of each outcome.
- Continuous probability distribution: is a set of possible range of values of a Continuous random variable along with the probability of the occurrence of this range of outcome.



Discrete probability distribution

It is a set of possible values of a discrete random variable along with the probability of the occurrence of each outcome.

There are three types of discrete probability distribution. They are

1. Binomial probability distribution(Bernoulli prob. Distribution)
2. Poisson probability distribution

3. Hyper geometric probability distribution

1. Binomial probability distribution

Characteristics

- Each trial (population) has only two possible outcomes. i.e. success or failure
- We are interested in the number of success in a fixed number of trials.
- The random variable counts the number of success in a fixed number of trials.
- All trials are independent
- The probability of success and failure remain the same for each trial.

🚩 Parameters(inputs) to construct Binomial probability distribution are

- Number of trials (n)
- The probability of success on each trial (P)

$$P(r) = p^r (q)^{(n-r)} C_r^n, r=0, 1, 2... n.$$

P(r) = Probability of r successes

r = exact number of successes

n = number of trials made

P = probability of success on one trial

q = probability of failure on one trial (q = 1-P)

$$C_n^N = \binom{N}{n} = \frac{N!}{n!(N-n)!}$$

🚩 For binomial distributions, the mean (also called the expected value of number of successes) and standard deviation is calculated by using a formula:

$$\mu = np$$

$$\sigma = \sqrt{npq}$$

Where: n = number of trials made

P = probability of success on one trial

q = probability of failure on one trial (q = 1-P)

σ = standard deviation

Example 1: A student doesn't know the answers to **five** multiple choice questions, each having four alternatives. He simply guesses the answers.

Required:

- A. Construct probability distribution for the number of correct responses.
- B. What is the probability of the student answering
 - i. Exactly 3 questions?
 - ii. No questions correctly?
 - iii. At least one question correctly?
 - iv. At least two questions correctly?
 - v. At least three questions correctly?
 - vi. At most three questions correctly?
- C. Calculate the mean and standard deviation of distribution.

Exercise1

Suppose 40% of the people entering a store make a purchase. If 10 people enter the store, find the expected number of people making a purchase?

Exercise 2

Car color preferences change over the years and according to the particular model that the customer selects. In a recent year, 10% of all luxury cars sold were black. If 25 cars of that year and type are randomly selected,

- A. Does the situation fulfill binomial experiment?
- B. Construct the appropriate probability distribution?
- C. Find the following probabilities
 - i. Exactly four cars are black?
 - ii. At least five cars are black?
 - iii. At most six cars are black?
 - iv. No more than 4 cars are black?
 - v. More than five cars are black?
 - vi. Between three and five (inclusive) cars are black?
- D. Calculate mean and standard deviation of the binomial distribution?

2. Hyper-geometric distribution

Characteristics

- The outcomes of each trial of an experiment are classified in to two group of outcomes; success and failure.
- The random variable is the number of success in a fixed number of trials.
- We assume sample from a finite population without replacement.

$$n/N \geq 0.05$$

- The trials are dependent.
- The probability of success changes from trial to trial

$$P(r) = \frac{C_r^R - C_{n-r}^{N-R}}{C_n^N}, r = 0, 1, 2, \dots, n.$$

Example 1

A store has 50 identical super xx tires in the stock. There are 10 tires slightly damaged steel mixed randomly in the stock. A customer buys a set of five tires.

Required: Construct probability distribution for the number of good (undamaged) tires obtained by the customers.

Example 2

NTA plant employs 40 peoples in the assembly department. 30 of the employees belong to a union and 10 do not. 4 employees are selected at random to form a committee that would discuss with the management. Suppose we want the number of union members for the selected ones.

Required:

- A. Does the experiment fulfill hyper geometric experiment?
- B. Construct the appropriate probability distribution?
- C. Find the following probabilities
 - i. At most two are the member of the union?
 - ii. At least three are member of the union?
 - iii. Not more than two are member of the union?

Exercise

A company has five applicants for two positions: two women and three men. Suppose that the five applicants are equally qualified and that no preference is given for choosing either gender. Let r equal the number of women choosing to fulfill the two positions.

Required: Construct probability distribution of r ?

3. *Poisson probability distribution*

It describes the number of occurrences of a specific event in a specified interval.

The interval may be time, distance, area, or volume.

Properties of a Poisson Experiment

- The probability of an occurrence is the same for any two intervals of equal length.
- The occurrence or non-occurrence in any interval is independent of the occurrence or non-occurrence in any other interval.
- It is extremely unlikely that, there will be more than one arrival (occurrence) in a very short interval.

$$P(x) = \frac{e^{-\mu} \mu^x}{x!}$$

Where

$P(x)$ = probability of x occurrences in an interval

μ = mean number of occurrences in an interval

$e = 2.71828$

x = the number of occurrences

- X is a discrete random variable that may assume an infinite sequence of values ($x = 0, 1, 2, 3, \dots$)

Example 1

Patients arrive at the emergency room of Mercy Hospital at the average rate of 6 per hour on weekend evenings.

Required:

- A. Check whether the properties of Poisson experiment are satisfied?
- B. Construct the appropriate probability distribution?
- C. What is the probability of
 - i. 4 arrivals in 30 minutes on a weekend evening?
 - ii. At least 3 arrivals in 60 minute on a weekend evening?
- D. Compute mean and standard deviation of the distribution?

Example 2

The average number of traffic accidents in Addis Ababa city is 2 per week. Required: Find the probability of

- A. No accident during the 1st week period?
- B. At most three accidents during a 2 week period?

Exercise 1

A machine has to be shut down for repairs on an average of two times in 24 days. When more than 4 shut downs occur in 24 days, the production schedule cannot be attained. If shut downs are a Poisson process, what is the probability that the production schedule cannot be attained?

BASIC REQUIREMENTS FOR ASSIGNING PROBABILITIES

1. The probability assigned to each experimental outcome must be between 0 and 1, inclusively.

If we let E_i denote the i th experimental outcome and $P(E_i)$ its probability, then this requirement can be written as

$$0 \leq P(E_i) \leq 1 \quad \text{for all } i$$

2. The sum of the probabilities for all the experimental outcomes must equal 1.0. For n experimental outcomes, this requirement can be written as

$$P(E_1) + P(E_2) + \dots + P(E_n) = 1$$